



Angles in a circle

THE ANGLES IN A CIRCLE HAVE A NUMBER OF SPECIAL PROPERTIES.

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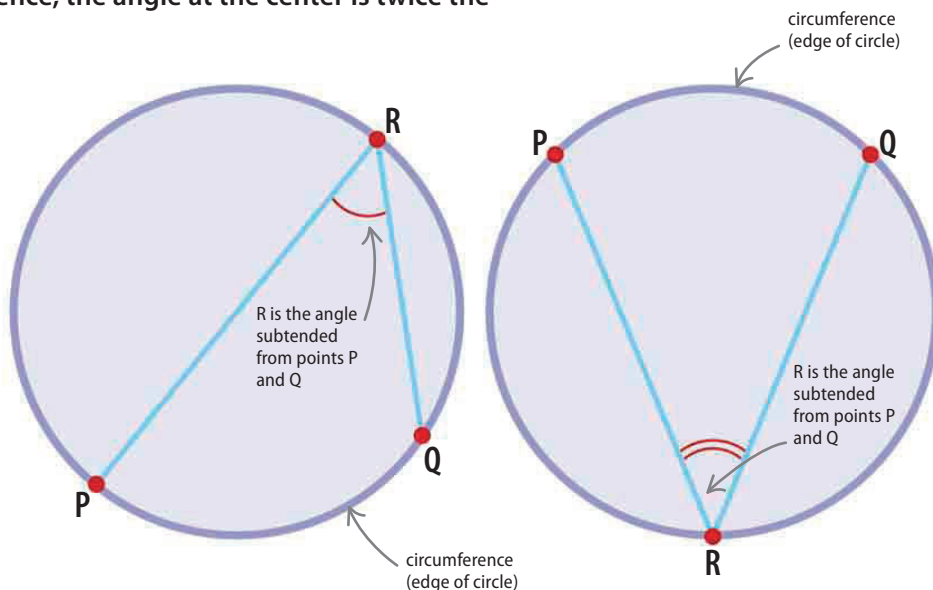
If angles are drawn to the center and the circumference from the same two points on the circumference, the angle at the center is twice the angle at the circumference.

Subtended angles

Any angle within a circle is “subtended” from two points on its circumference—it “stands” on the two points. In both of these examples, the angle at point R is the angle subtended, or standing on, points P and Q. Subtended angles can sit anywhere within the circle.

▷ Subtended angles

These circles show how a point is subtended from two other points on the circle’s circumference to form an angle. The angle at point R is subtended from points P and Q.



Angles at the center and at the circumference

When angles are subtended from the same two points to both the center of the circle and to its circumference, the angle at the center is always twice the size of the angle formed at the circumference. In this example, both angles R at the circumference and O at the center are subtended from the same points, P and Q.

$$\text{angle at center} = 2 \times \text{angle at circumference}$$

▷ Angle property

The angles at O and R are both subtended by the points P and Q at the circumference. This means that the angle at O is twice the size of the angle at R.

